

Revisiting Fundamentals of the European Gas Market

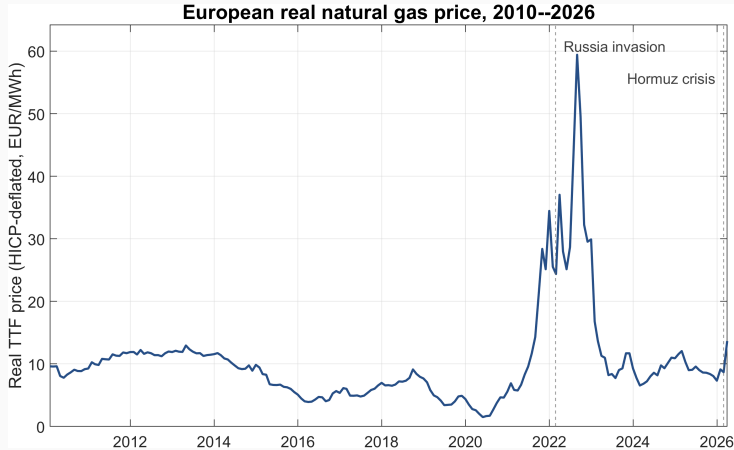
The Role of Supply Substitution

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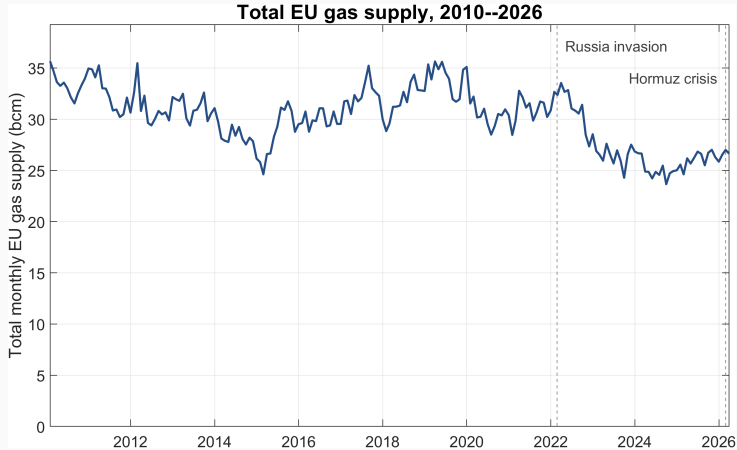
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An unprecedented price episode



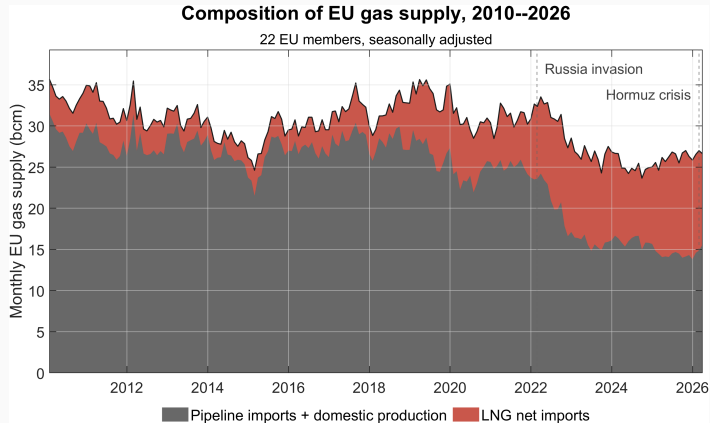
x: monthly time 2010–2026. y: real Dutch TTF price (HICP-deflated, EUR/MWh). Dashed lines mark the Feb-2022 Russia invasion and the Feb-2026 Hormuz crisis.

Total EU gas supply



x: monthly time 2010–2026. y: total monthly EU gas supply (bcm) = pipeline imports + domestic production + LNG net imports, summed across 22 EU members.

...decomposed: pipeline-and-domestic vs LNG



*Same total as the previous slide, split by source. Pipeline-and-domestic (grey) roughly **halves** between 2021 and 2024; LNG (red) **nearly triples** to partly fill the gap.*

See pipeline and LNG separately

The gap in the literature.

- Existing SVAR work on the natural gas market treats supply as a single aggregate.
- Pipeline and LNG differ in short-run elasticity by an order of magnitude.
- Aggregating the two channels understates supply-driven price dynamics.

This paper.

- First Bayesian SVAR of the EU gas market with explicit pipeline / LNG split.
- Identification via sign + elasticity restrictions (Baumeister–Hamilton, 2015, 2019).
- Three retrospective counterfactuals for the 2022 crisis + a forward-looking scenario fan for the ongoing 2026 Hormuz disruption.

Are all supply shocks alike?

– *i.e.*, are the 2022 Russian pipeline shock and the 2026 Hormuz LNG shock economically equivalent disturbances, or do they transmit to prices through structurally different channels?

Three sub-questions:

1. How different are the **feedback effects** of each type of supply disruption?
2. How much of the 2022 spike was **disruption**, how much **LNG bottleneck**, how much **demand savings**?
3. Under the 2026 **Strait of Hormuz** disruption, how does the price response depend on the resolution path?

Preview of findings

- **Elasticities.** LNG is **much more responsive** to price than pipeline ($\beta_{qp} \approx 1.6$ vs $\gamma_{qp} \approx 0.08$).
- **IRFs.** New evidence on supply heterogeneity: pipeline shocks transmit as **transitory** price spikes (~ 5 months), LNG shocks as **persistent** medium-run shifts in the price level.
- **Three retrospective counterfactuals for the 2022 crisis:**
 - CF1 — LNG couldn't react in the 2022: **+0.80 log-points** ($\approx 2.2\times$ realised).
 - CF2 — no regasification bottleneck: **-1.70 log-points** ($\approx 0.18\times$ realised).
 - CF3 — no autonomous demand savings: **+0.30 log-points** ($\approx 1.35\times$).

$\Rightarrow$ realised path **closer to CF1 than to CF3**; the bottleneck binds.
- **2026 Hormuz fan.** 12-month price gap ranges from **-5% (prompt recovery)** to **+127% (escalation)**; observed May-2026 evidence sits closer to the persistent scenario.

Variables (monthly, Feb 2010 – Mar 2026):

$$y_t = [q_t^{\text{pip}}, ip_t, p_t, s_t, q_t^{\text{LNG}}]'$$

- q^{pip} : pipeline imports + domestic production, 22 EU members (JODI).
- ip : euro-area industrial production (Eurostat).
- p : real Dutch TTF price (World Bank, HICP-deflated).
- s : total gas inventories (JODI).
- q^{LNG} : net LNG imports, 22 EU members (JODI).

SVAR specification

Five structural equations (lag terms suppressed)

$$\begin{aligned}q_t^{pip} &= \gamma_{qp} p_t + \varepsilon_t^{\text{pip sup}} \\q_t^{LNG} &= \beta_{qp} p_t + \varepsilon_t^{\text{LNG sup}} \\ip_t &= \alpha_{yp} p_t + \varepsilon_t^{\text{agg dem}} \\Q_t &= \alpha_{qy} ip_t + \alpha_{qp} p_t + \varepsilon_t^{\text{flow dem}} \\s_t &= \psi_1 q_t^{pip} + \psi_2 p_t + \psi_3 q_t^{LNG} + \varepsilon_t^{\text{prec dem}}\end{aligned}$$

Gas-market clearing identity $Q_t = q_t^{pip} + q_t^{LNG} - s_t$.

Restrictions implemented as truncated Student- t priors on A (Baumeister–Hamilton, 2015, 2019); $p = 6$ lags; 2×10^6 -draw MCMC, 3×10^5 retained.

Sign restrictions

$$\begin{aligned}\gamma_{qp} &> 0 && \text{pipe supply} \\ \beta_{qp} &> 0 && \text{LNG supply} \\ \alpha_{yp} &< 0 && \text{price} \rightarrow \text{IP} \\ \alpha_{qy} &> 0 && \text{income} \rightarrow \text{demand} \\ \alpha_{qp} &< 0 && \text{demand slope} \\ \psi_1, \psi_2, \psi_3 &&& \text{none}\end{aligned}$$

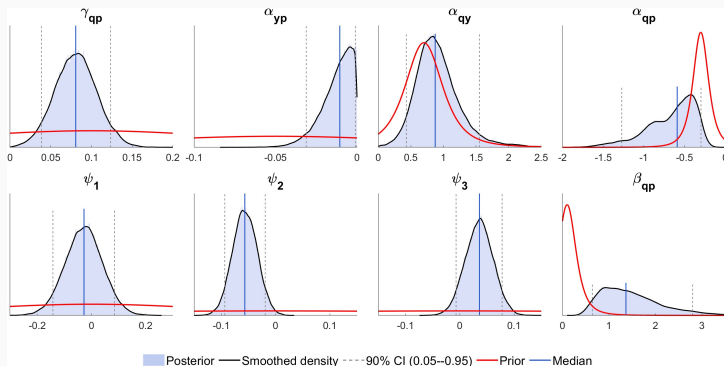
Full restrictions

Posterior formulas

MCMC diagnostics

Structural shocks

Posterior contemporaneous elasticities



Each panel: x = parameter value; y = posterior density. Solid = posterior, dashed = prior.

Posterior median

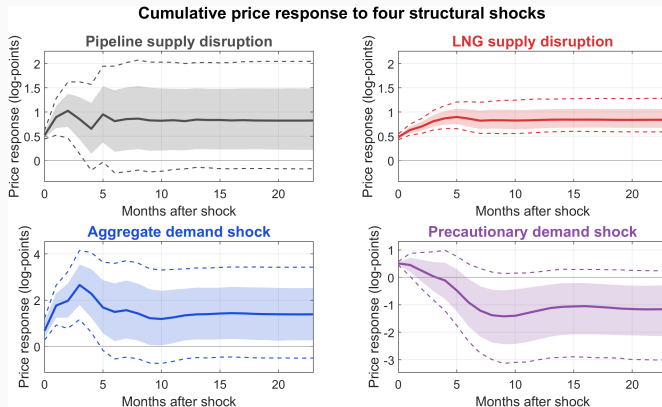
- LNG supply slope
 $\beta_{qp} \approx 1.6$ (flexible)
- Pipeline supply slope
 $\gamma_{qp} \approx 0.08$ (rigid)
- Demand slope
 $\alpha_{qp} \approx -0.53$
- Inventory price elast.
 $\psi_2 \approx -0.07$

Marginals of A

A^{-1} impact

Priors & posteriors table

Impulse responses: price effects of four structural shocks



Cumulative price response over 24 months. Solid = posterior median; shaded band = 68% credible interval; dashed lines = 90%. Y-axis scaled per panel. Supply shocks sign-flipped to read as disruptions.

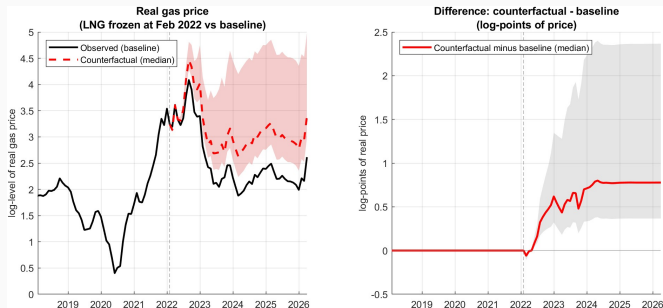
Reading

- Both supply disruptions raise prices; LNG effect is permanent.
- Aggregate demand dominates in magnitude, persistent.
- Precautionary shock: initial spike, then inventory release pulls price below baseline.

Full 5 × 5 IRF grid

Historical decomposition

CF1: what if LNG had not expanded in 2022?



Left: real gas price (black = observed; red dashed = counterfactual median with 5–95% band). Right: counterfactual–baseline gap in log-points of real price. Vertical dashed line = start of the crisis window (Feb 2022).

Counterfactual: q_t^{LNG} frozen at its Feb-2022 level over the EU crisis window.

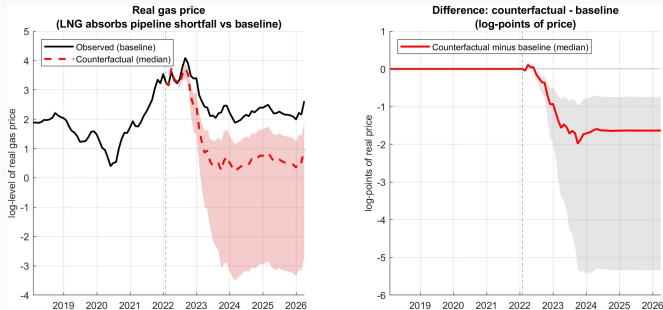
Result. Median price gap $\hat{\Delta}p_{\text{Dec-23}} \approx +0.80 \text{ lp} \Rightarrow$ prices would have been $\sim 2.2\times$ realised.

Reading. European LNG absorbed about half of the disruption.

Full multi-variable response

LNG-trend variant

CF2: what if there had been no regasification bottleneck?



Same layout as CF1. Counterfactual = LNG fully absorbs contemporaneous pipeline shortfall (Europe unconstrained on the receiving side).

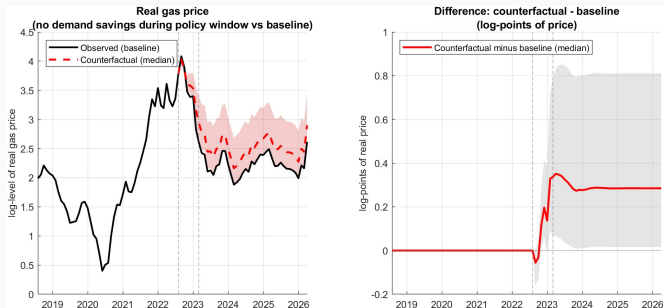
Counterfactual: LNG *fully* absorbs the contemporaneous pipeline-and-domestic shortfall (Europe unconstrained on the receiving side).

Result. $\hat{\Delta}p \approx -1.70 \text{ lp} \Rightarrow$ prices would have been $\sim 18\%$ of realised.

Asymmetry. Realised path *closer to CF1 (frozen)* than to CF2 (unconstrained) $\Rightarrow$ regasification *bottleneck binds*.

Full multi-variable response

CF3: what if the EU 15% demand-savings target hadn't kicked in?



Same layout. Counterfactual = autonomous demand-saving shock u_t^3 set to zero over the policy window.

Counterfactual: u_t^3 (precautionary / savings demand shock) set to zero over the EU Council Regulation 2022/1369 policy window.

Result. $\hat{\Delta}p \approx +0.30$ lp $\Rightarrow$ prices $\sim 35\%$ higher than realised.

Reading. Material but **smaller than the supply margin**. Substitution was the dominant absorption channel.

Full multi-variable response

CF4: how the three scenarios are built

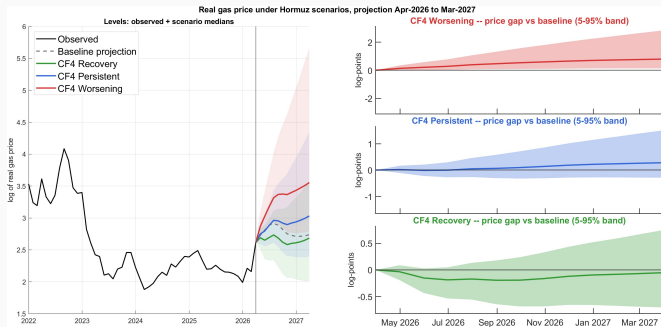
Anchor. The realised March-2026 LNG log-growth $\Delta \equiv \Delta_{\text{Mar-26}}^{\text{LNG}} \approx -9.3$ pp (first full month of the Hormuz disruption, already in the sample).

Imposed paths on monthly LNG log-growth g_t^{LNG} over April 2026 – March 2027:

Scenario	Story	Imposed g_t^{LNG}
CF4a Recovery	Rapid de-escalation; cargo flows resume	$-\Delta/2$ in Apr–May, then 0
CF4b Persistent	Chronic instability; redirection to Asia	0 throughout
CF4c Worsening	Second wave + spillover to other chokepoints	Δ in Apr, $\Delta/2$ in May, then 0

Mechanism. For each posterior draw, the LNG structural shock u_t^{LNG} is set so that g_t^{LNG} matches the scenario path; all other shocks are 0 over the projection horizon (deterministic-projection convention, Antolín-Díaz, Petrella & Rubio-Ramírez 2021). Bands reflect *parameter* uncertainty under the imposed LNG path, not future non-LNG shocks.

CF4: a forward-looking scenario fan for the 2026 Hormuz crisis



Left: log real gas price level. Black solid = observed through Mar 2026; dashed grey = unconstrained baseline projection; coloured = scenario medians with 5–95% bands. Right (stacked): per-scenario price gap vs baseline in log-points.

Three twelve-month scenarios, anchored on the realised March 2026 LNG drop (−9.3 pp):

Scenario	Gap (lp)	Multiple
CF4a Recovery	−0.05	0.95×
CF4b Persistent	+0.30	1.34×
CF4c Worsening	+0.82	2.27×

Observed May 2026 path lies between CF4b and CF4c, closer to CF4b.

Full multi-variable response

Take-aways

1. Supply contributions have historically been underestimated because of misspecification.
2. In 2022, LNG absorbed about half of the disruption; had it not, prices would have been $\sim 2.2\times$ the realised path.
3. The regasification bottleneck binds: realised prices sit close to the LNG-frozen upper bound, far from the unconstrained lower bound.
4. Autonomous demand savings contributed $\sim 35\%$ of price stabilisation – material but secondary to substitution.
5. 2026 Hormuz: even a worsening scenario is smaller in magnitude than 2022; the asymmetric fan implies receiving-side infrastructure and supplier diversification are complementary energy-security instruments.

Thank you!

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Appendix

Data exhibits

Identification details

Historical decomposition

Full IRF grid

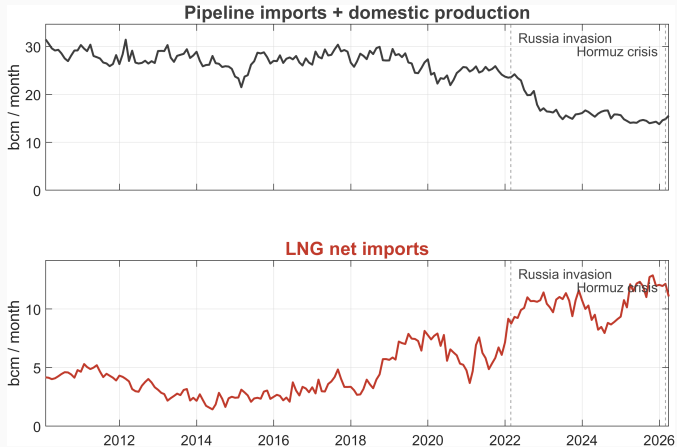
Counterfactuals: full multi-variable response

CF1b: LNG-trend variant

Robustness: IEA Gas Trade Flows

Data exhibits

EU gas supply: pipeline and LNG separately



Top: pipeline imports + domestic production (grey, ~30 bcm/m pre-2022, ~15 bcm/m post-2024). Bottom: LNG net imports (red, ~4 bcm/m pre-2020, ~11 bcm/m post-2022).

Identification details

Sign and elasticity restrictions on A

Structural form $Ay_t = B(L)y_{t-1} + u_t$, with $u_t \sim \mathcal{N}(0, \mathbf{D})$. Five contemporaneous elasticities carry a sign restriction; three storage parameters are left unrestricted.

Parameter	Meaning	Sign restriction
γ_{qp}	Pipeline supply elasticity	$\gamma_{qp} > 0$
β_{qp}	LNG supply elasticity	$\beta_{qp} > 0$
α_{yp}	Effect of p on economic activity	$\alpha_{yp} < 0$
α_{qy}	Income elasticity of gas demand	$\alpha_{qy} > 0$
α_{qp}	Gas demand elasticity	$\alpha_{qp} < 0$
ψ_1	Effect of q^{pip} on gas inventories	none
ψ_2	Effect of p on gas inventories	none
ψ_3	Effect of q^{LNG} on gas inventories	none

Sign restrictions are enforced through truncated Student- t priors on the elements of A , centred on literature-based values (Baumeister & Hamilton, 2015, 2019).

Posterior formulas (Theorem 1, BH 2015)

Joint posterior decomposition.

$$p(A, B, D \mid Y_T) = p(A \mid Y_T) p(D \mid A, Y_T) p(B \mid A, D, Y_T)$$

Posterior of D (diagonal covariance, inverse Gamma):

$$d_{ii}^{-1} \mid A, Y_T \sim \Gamma(\kappa_i^*, \tau_i^*(A)), \quad \kappa_i^* = \kappa_i + \frac{T}{2}, \quad \tau_i^*(A) = \tau_i + \xi_i^*(A)$$

Posterior of B (VAR coefficients, conditional normal):

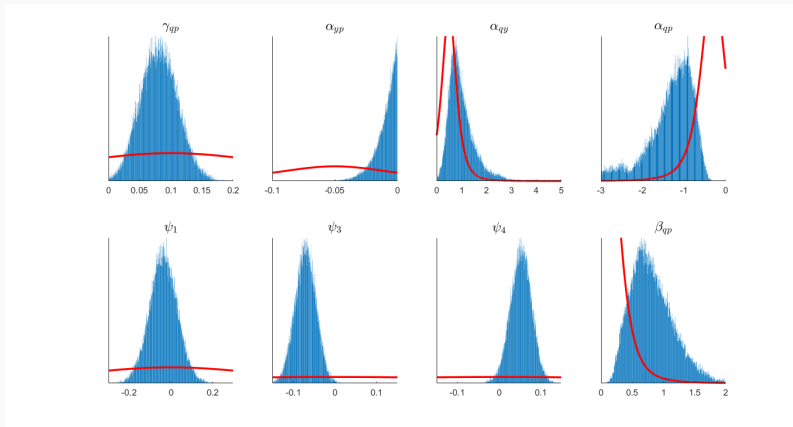
$$b_i \mid A, D, Y_T \sim \mathcal{N}(m_i^*, d_{ii} M_i^*), \quad m_i^* = (\tilde{X}_i' \tilde{X}_i)^{-1} \tilde{X}_i' \tilde{Y}_i, \quad M_i^* = (\tilde{X}_i' \tilde{X}_i)^{-1}$$

Posterior of A (structural matrix, sampled via Metropolis–Hastings):

$$p(A \mid Y_T) = k_T p(A) |\det(A \hat{\Omega}_T A')|^{-T/2} \prod_{i=1}^n \left(\frac{2\tau_i^*}{T} \right)^{-\kappa_i^*} \prod_{i=1}^n \frac{|M_i^*|^{1/2}}{|M_i|^{1/2}} \frac{\tau_i^{\kappa_i} \Gamma(\kappa_i^*)}{\Gamma(\kappa_i)}$$

D and B admit closed-form conditional posteriors; A does not, hence Metropolis–Hastings step.

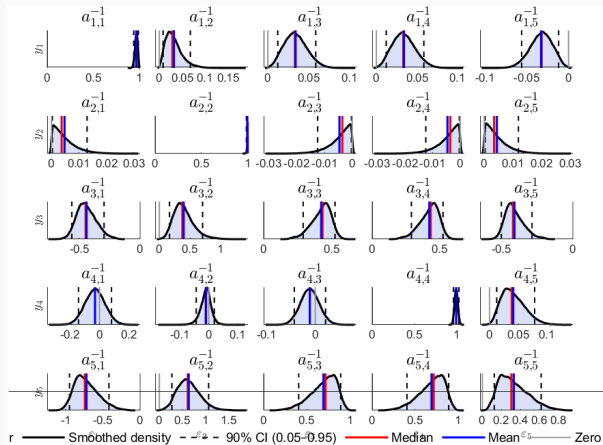
Posterior of A — marginal densities



x : parameter value; y : posterior density (Kernel smoothed). Vertical line = posterior median.

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Posterior of A^{-1} — implied impact responses



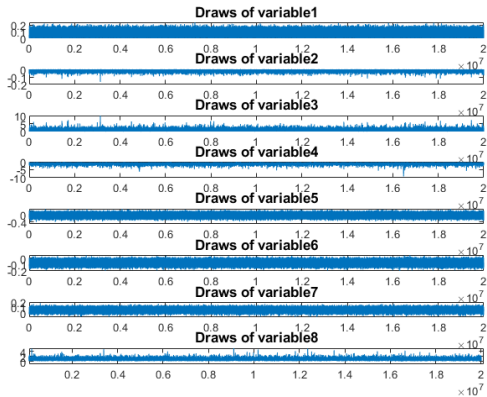
x : contemporaneous impact of a unit structural shock; y : posterior density.

Priors and posteriors for the contemporaneous matrix A

	γ_{qp}	β_{qp}	α_{yp}	α_{qy}	α_{qp}	ψ_1	ψ_2	ψ_3
<i>Priors (truncated Student-t)</i>								
Location	0.1	0.1	-0.05	0.7	-0.3	0	0	0
Scale	0.2	0.2	0.1	0.3	0.1	0.5	0.5	0.5
Degrees of freedom	3	3	3	3	3	3	3	3
Sign restriction	t^+	t^+	t^-	t^+	t^-	t	t	t
<i>Posterior</i>								
5%	0.034	0.825	-0.028	0.446	-1.098	-0.157	-0.108	-0.011
50%	0.079	1.629	-0.009	0.894	-0.530	-0.041	-0.067	0.030
95%	0.125	3.028	-0.001	1.535	-0.283	0.073	-0.026	0.072
Mean	0.079	1.745	-0.011	0.929	-0.592	-0.042	-0.067	0.030

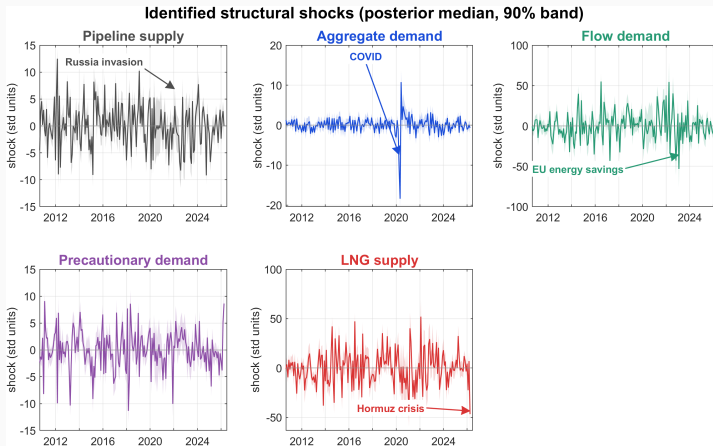
Posterior estimates from 300,000 MCMC draws after burn-in. t^+/t^- denote truncation to the positive / negative orthant.

MCMC convergence



x: MCMC draw index; y: running mean of selected A entry. Stationarity & mixing across multiple chains.

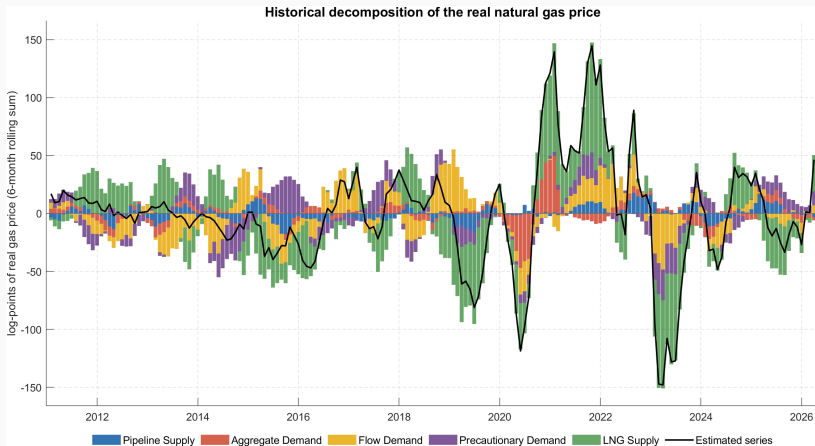
Identified structural shocks



x : monthly time 2010–2026. y : estimated structural shock ε_t (posterior median, 90% credible band). Note the COVID spike in aggregate demand and the post-2022 volatility in LNG supply.

Historical decomposition

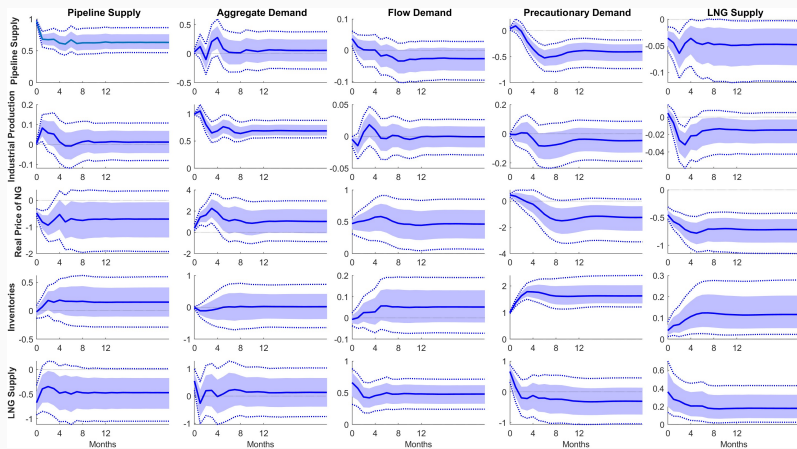
Historical decomposition of the real gas price



x: monthly time 2010–2026; y: log-points of the real TTF price. Stacked bars = cumulative six-month contribution of each structural shock.

Full IRF grid

Cumulative IRFs to all five structural shocks

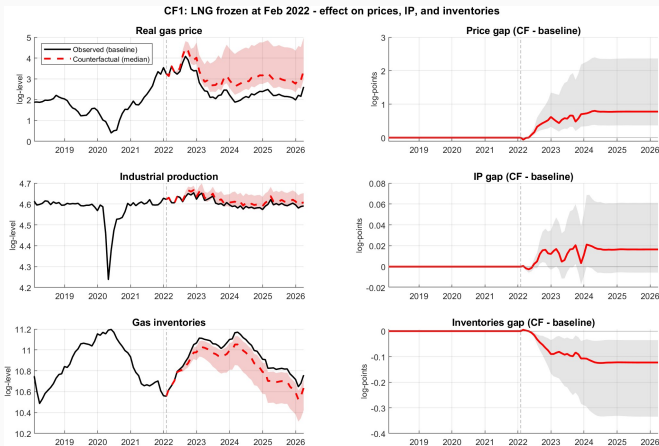


Rows: response variable. Columns: structural shock. x: months 0–24; y: cumulative log-point response.

Shaded: 5–95% posterior credible band.

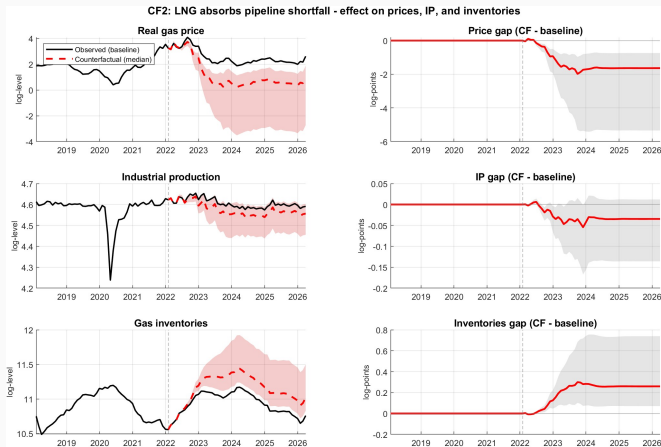
Counterfactuals: full multi-variable response

CF1: full multi-variable response (LNG frozen)



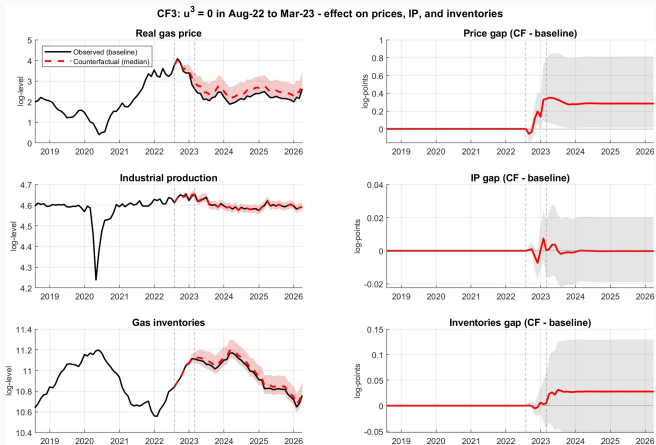
5-panel grid (one per variable); x = months 2022–2024; y = log-level path. Black solid = observed; coloured = counterfactual posterior median with 5–95% band.

CF2: full multi-variable response (no bottleneck)



Same layout as CF1. Counterfactual = LNG fully absorbs contemporaneous pipeline shortfall.

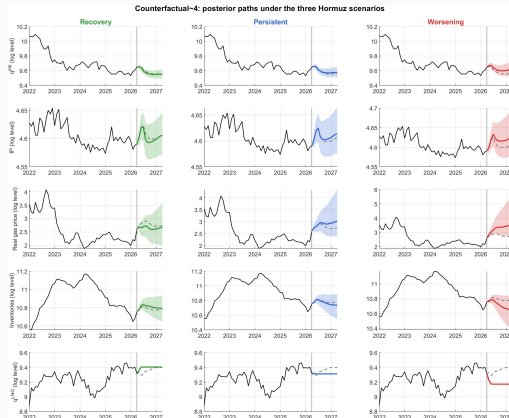
CF3: full multi-variable response (no demand savings)



Same layout. Counterfactual = autonomous demand-saving shock u_t^3 set to zero over the policy window.

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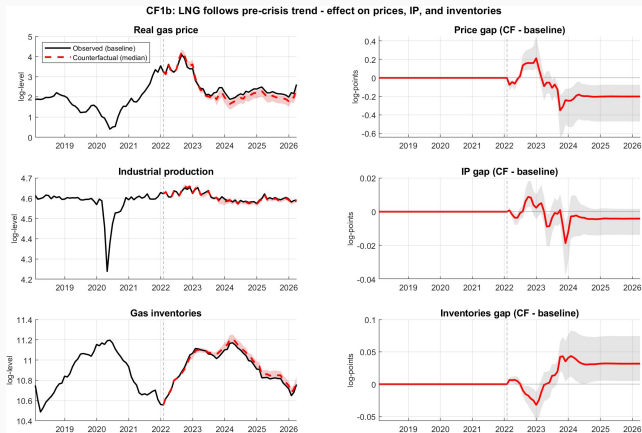
CF4: full multi-variable response (Hormuz fan)



Rows: the five endogenous variables (q^{pP} , ip , p , s , q^{LNG}). Columns: the three Hormuz scenarios (Recovery, Persistent, Worsening). Solid black = observed, dashed grey = unconstrained baseline, coloured = scenario median + 5–95% band.

CF1b: LNG-trend variant

CF1b: LNG continues its pre-crisis trend



Same layout as CF1. Counterfactual = q^{LNG} follows its 2018–2021 trend instead of the sharp 2022 acceleration.

Less extreme than CF1: LNG still expands, but at trend speed. Median gap small and centred near zero — the trend path nearly coincides with the realised path at the end of the crisis window.

Robustness: IEA Gas Trade Flows

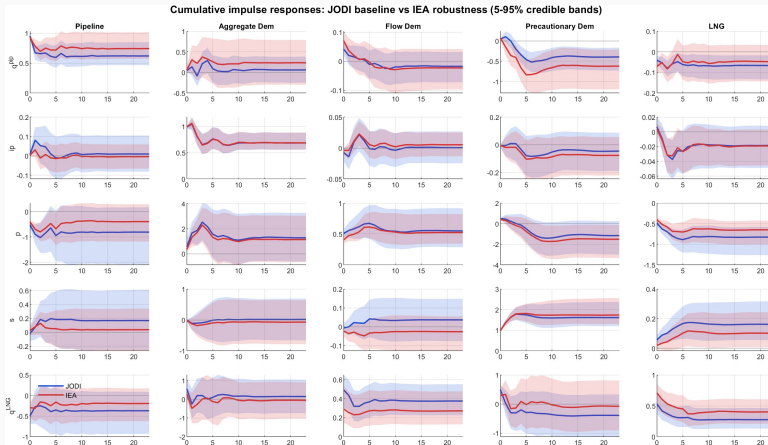
Same identification, alternative supply data

Parameter	JODI baseline	IEA robustness
β_{qp} (LNG supply on price)	+1.365 [+0.64, +2.80]	+0.608 [+0.25, +1.17]
γ_{qp} (pipe supply on price)	+0.081 [+0.04, +0.12]	+0.176 [+0.11, +0.25]
α_{qp} (demand on price)	+0.590 [+0.30, +1.27]	+1.702 [+0.96, +3.28]
α_{qy} (demand on IP)	+0.874 [+0.43, +1.55]	+0.849 [+0.37, +1.65]
ψ_2 (inventory on price)	-0.057 [-0.09, -0.02]	-0.047 [-0.09, -0.01]
α_{yp} (IP on price)	-0.011 [-0.03, -0.00]	-0.010 [-0.03, -0.00]

All signs preserved. The flexible-rigid ranking remains (LNG more elastic than pipeline), with a compressed ratio ($\sim 3.4\times$ vs $\sim 20\times$).

[IRF comparison](#)[Data comparison](#)[◀ Back](#)

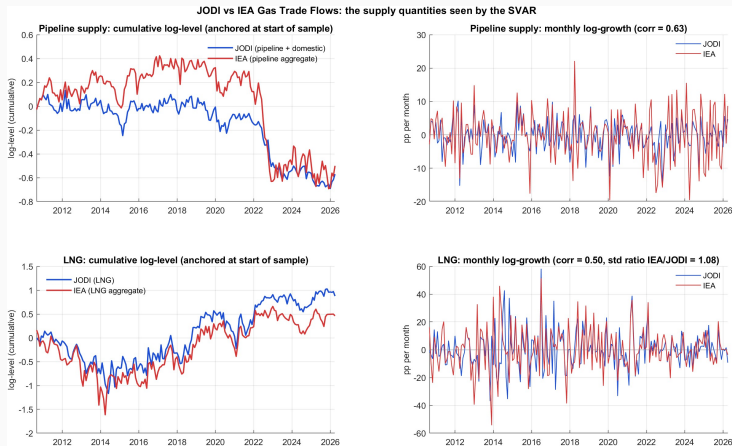
IRF comparison: JODI baseline vs IEA robustness



Rows = response, columns = shock. x: months 0–24; y: cumulative log-point response.

Blue: JODI. Red: IEA. Bands: 5–95%.

Data comparison: JODI vs IEA supply aggregates



Top row: pipeline supply. Bottom row: LNG. Left: cumulative log-level (anchored at sample start). Right: monthly log-growth.

References i

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